Investigation 5: Chromatography a bonding competition

Chromatography is a laboratory technique used to separate mixtures. A mixture dissolved in a solvent called the *mobile phase* is passed through a *stationary phase*, like paper or silica gel. It works because the mobile phase, stationary phase and mixture compound have different polarities. These differences in polarity mean differences in the attraction of mixture components to the stationary phase and in their solubility in the solvent. This results in their separation as they travel through the stationary phase at different rates.

In paper chromatography for example, the paper is made of cellulose, a polar substance. The more polar compounds within a mixture being separated will bond more readily with the polar cellulose paper. As a result the components become separated. The more polar the component the less they travel up the paper while the more non-polar components travel further up the paper.

Separation of compounds using chromatography is therefore based on a competition. The competition is for bonding places on the stationary phase. The compounds in the sample and the mobile phase compete for bonding places on the stationary phase.

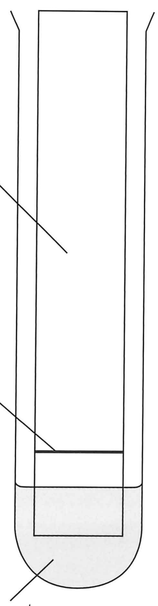
Notes

The task

Identify the most effective solvent, from a selection of solvents, for the separation of the coloured dies from black ink using paper chromatography.

Equipment

test tubes (five large)
strips of chromatography or filter paper to fit into the large test tubes (five)
household solvents - 5 mL samples:
sodium chloride solution [NaCℓ] 1% solution
cloudy ammonia solution
methylated spirits
tap water
kerosene
black ink: use ink from a black felt tipped pen or ballpoint pen



Planning the investigation

1. Plan an investigation using paper chromatography to determine the most effective solvent, mobile phase, to use to separate the coloured dies from black ink. In your plan include a table of the variables considered in the investigation.

	Variable/s	Unit/s	How the variable will be measured/ manipulated
Independent variable			
Dependent variable			
Factors kept constant (controlled variables)			

2. Briefly outline your plan. A list of equipment and solvents and a sample setup has been provided but with your teacher's permission alternatives may be used. List the chemicals and equipment you need and identify safety requirements.

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Notes

3 Check the proposed procedure with your teacher.

SAFETY NOTE:
- **Check your plan with your teacher before you commence.**
- **At the end of the investigation dispose of the solvents appropriately – method of solvent disposal must be included in your procedure.**

Conducting the investigation

1. Conduct some preliminary trials. You may need to clarify how the independent and dependent variables will be measured.

2. Describe what you learned from these initial trials and modifications that you made to your original plan.

3. Conduct the investigation, collecting and recording the data in a table as you proceed. If time allows replicate the data collection and average the results if appropriate.

Processing the results

Analyse your data and relate your conclusions to the original purpose of the investigation. Include in your discussion:

- The different coloured components in black ink and their relative polarities.
- Competition of the coloured components for binding positions on the polar cellulose paper, i.e the stationary phase.
- Comparison of the solubilities of the different coloured components with each of the solvents.

Conclusion

Which is the most effective solvent, for the separation of the coloured dies from black ink using paper chromatography and why?

Evaluating the investigation

1. Evaluate the effectiveness of your procedure and describe any modifications you would make to improve it. You may discuss factors that would improve the accuracy of your results such as sample size and selection, measurement errors and the control of variables. As well, you may address more general organisational factors such as the allocation of tasks among group members and the nature of the apparatus and how it was set up.

2. Discuss your confidence in the findings of the investigation.

3. Recommend procedures that can be carried out and investigations that need to be done to further improve the effectiveness of the solvent you have identified as the best to use for the separation of black ink.